Globally Stable I/O Robust Control of Flexible Joint Robots

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Abstract

This paper presents the first input-output robust control design for the trajectory following problem of a flexible-joint robot manipulator. The proposed design provides a class of controllers which only require position and velocity feedback and ensure global stability. The resulting stability is that the tracking error can be made to be smaller than a design parameter arbitrarily chosen by the designer. The proposed control is robust since it guarantees tracking performance in the presence of high-order nonlinear uncertainties including unknown joint elasticity, unknown parameters, load variation, disturbances. It is practically more important that no measurement on acceleration, jerk, or position and velocity deformation is required.

1 Introduction

Control design for robot manipulators with joint flexibility has attracted much attention from control researchers in recent years. The main reason is that, as shown by experimental study in [21], joint flexibility must be taken into account in both modelling and control design in order to achieve high tracking performance. Common sources of joint flexibility are gear elasticity (for example, harmonic drives), shaft windup, etc. Most researchers have adopted the model of flexible joint robots presented in [17].

Several methods, such as feedback linearization, observer design, adaptive control, singular perturbation, and robust control, have been investigated to design effective control for flexible joint robots. A list of references on these results can be found in the survey paper [19], we shall only brief synopsis of the most recent developments and provide comparison of different approaches. In feedback linearization method [17, 8], control is designed based on feedback linearization transformation which requires the knowledge of the dynamics and acceleration and jerk measurements. Under the assumption that joint flexibility is "small", there are two time scales in system dynamics, and a control can be designed using singular perturbation technique. Adaptive version of this slow/fast control was studied in [5, 6]. The resulting control is intuitively simple since it usually contains two parts: a control for rigid body and a corrective control, however, the control does not guarantee global stability. The most recent adaptive control scheme proposed in [9] guarantees global stability under the standard assumption that dynamics can be fully parameterized. On the front of robust control design, local stability was shown [16] which can be viewed as extension of robust control results for rigid robots [11, 13]. Later, a robust control guaranteeing global stability was presented in [3], this full state feedback robust control design is an application of the result in [15] and an extension of the robust control law [2] for rigid robots. Hybrid control, or combined robust and adaptive control, of flexible joint robots has also been studied [12, 4].

So far all existing control laws for elastic robots require full state feedback. There are some results [10, 22] on designing state observer for flexible joint robots, but no result has been reported on the subject of how to close the control loop, that is, how to design a control based on the estimates of the states. Since the separation principle does not hold in general for nonlinear systems, it appears unlikely that full state feedback control can be changed to an estimate-based control to guarantee global stability.

In the present paper we consider for the first time the input-output control problem of flexible joint manipulators. The control objective is to design a robust control which guarantees global stability and good tracking performance but requires only feedback of link position and velocity. The proposed design method is inspired by the recent result [14] on robust input-output design of mainly linear systems. The proposed control design is not only more attractive in practice since it requires less feedback information, but also allows the presence of significant but bounded nonlinear uncertainties including fast time varying parameters, modelling errors, load change, and unknown flexibility. It is our intention in this paper to propose

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the new design technique and to develop rigorous stability analysis. Simulation and experimentation will be performed in a near future.

The paper is organized as follows. In section 2, dynamics of flexible joint robots and its properties are presented, and necessary assumptions are introduced. In section 3, we first detail the input-output robust control design for the case that flexibility and its damping are constant and known. This relatively simple setting allows us to reveal the essential features of input-output robust control design and to proceed theoretical proof without too much complications. Then, the general case that flexibility and damping are unknown (and possible time-varying) is investigated in section 4 in which robust control is redesigned and the same stability results follow. Finally, some conclusions are made in section 5.

2 Problem Formulation

We shall consider the dynamics of a robot with flexible joints to be described by the following nonlinear differential equations:

$$\begin{aligned} 0 &= M(q_1)\ddot{q}_1 + N(q_1,\dot{q}_1) + K(q_1 - q_2) \quad (1)\\ J\ddot{q}_2 &= K(q_1 - q_2) - D\dot{q}_2 + \tau + P(q_1,\dot{q}_1) \quad (2)\\ N(q_1,\dot{q}_1) &= V_m(q_1,\dot{q}_1)\dot{q}_1 + G(q_1) + F_d\dot{q}_1 + H(q_1,\dot{q}_1)\\ H(q_1,\dot{q}_1) &= F_s(\dot{q}_1) + T_d, \end{aligned}$$

where q_1 and q_2 represent $n \times 1$ vectors of joint angles and motor angles, respectively, τ is $n \times 1$ the control vector of motor output torques, $M(q_1)$ is an $n \times n$ inertia matrix(symmetric and positive definite) for the rigid links, K is a diagonal matrix representing the joint stiffness. $V_m(q_1,\dot{q}_1)$ is an $n \times n$ matrix of centripetal and Coriolis terms, $G(q_1)$ is an $n \times 1$ vector of gravity terms, F_d is an $n \times n$ diagonal matrix of dynamic friction coefficients, $H(q_1,\dot{q}_1)$ includes static friction $F_s(\dot{q}_1)$ and any unknown but bounded disturbances. It is worth noting that T_d is in general a ***unknown*** nonlinear functional of q_1, $\dot{q}_1$. In equation (2), J is a diagonal matrix of actuator inertias reflected to the link side of the gears, D is also a diagonal matrix of torsional damping coefficients, $P(q_1,\dot{q}_1)$ denotes any possible nonlinearity whose bounding function depends only on q_1, and $\dot{q}_1$.

Let q_1^d characterize the desired, smooth trajectory that the robot should track. The assumption of q_1^d being smooth implies that q_1^d and its derivatives up to the forth order are continuous and bounded functions of time. It is apparent that this assumption is naturally satisfied in practice.

The followings are the important properties of robot dynamics that will be used in this paper.

P.1 It has been shown in [1] that the inertia matrix satisfies inequality $\underline{m}I \le M(q_1) \le \overline{m}(q_1)I$ for all $q_1 \in \mathbf{R}^{n\times 1}$, where it is assumed that $\underline{m}$ and $\overline{m}(q_1)$ be known. Note that $\overline{m}(q_1)$ is a non-negative function and reduces to a constant for robots with only revolute joints.

P.2 It has been shown in [1] that $V_m(q_1,\dot{q}_1)$ is a function at most of first order in q_1 and $\dot{q}_1$. Therefore, $\|V_m(q_1,\dot{q}_1)\| \le \beta_1 + \beta_2\|x_1\| = \rho_1(x_1)$, where $\|\cdot\|$ denotes Euclidean norm, x_1 is the ***output vector*** defined by $x_1 = \begin{bmatrix} e_1^T & \dot{e}_1^T \end{bmatrix}^T$, and e_1 is an $n \times 1$ vector representing the trajectory error, i.e. $e_1 = q_1^d - q_1$.

The robust control design proposed in this paper requires only the following assumptions:

A.1 *Unknown* functionals in dynamics equations (1) and (2) include dynamic and static friction, bounded disturbances, etc. It is assumed that they be bounded by known functions as: $\|G(q_1) + F_d\dot{q}_1 + F_s(\dot{q})\| \le \beta_3 + \beta_4\|x_1\| = \rho_2(x_1)$, $\|T_d\| \le \rho_3(x_1,t)$, $\|P(q_1,\dot{q}_1)\| \le \rho_4(x_1)$, and $\|M(q_1)\| \le \rho_5(x_1)$.

A.2 The feedback information for control design are only the measurements of joint position and velocity, i.e., q_1 and $\dot{q}_1$. The position and velocity deformations $q_2 - q_1$ and $\dot{q}_2 - \dot{q}_1$ are unknown, but initial deformations are bounded, that is, $\|x_2(t_0) - x_1(t_0)\| \le \rho_6$, or $\|x_2(t_0)\| \le \rho_7$ for some constant ρ_6 or ρ_7, where x_2 is the internal state defined by $x_2 = \begin{bmatrix} q_2^T & \dot{q}_2^T \end{bmatrix}^T$.

A.3 If the parameters K, J, and D are unknown, it is assumed that their value are limited within compact sets as $0 < \underline{K} \le K \le \overline{K}$, $0 < \underline{J} \le J \le \overline{J}$, and $0 < \underline{D} \le D \le \overline{D}$. In addition, if they are time-varying, their derivatives up to third order are bounded.

The control problem addressed in this paper is to develop an input-output robust control design procedure for flexible joint robots. The main difference between this work and previous work on robust control in [3, 4] is that the control to be designed require only the input and output feedback information, i.e., x_1 and τ. Measurement of acceleration, or jerk, or deformation is not required. The control is called to be robust since it requires no exactly knowledge of the systems, that is, nothing beyond the properties listed above. The control objective is to make the robot track any given trajectory with arbitrarily selected accuracy despite of unknown but bounded nonlinear dynamics.

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We can rewrite system dynamics (2) into the following state space representation:

$$\dot{x}_2 = A_2 x_2 + B_2[\tau + Kq_1 + P(q_1, \dot{q}_1)], q_2 = C_2 x_2 \quad (3)$$

where $q_2 = C_2 x_2$, $C_2 = \left[\begin{array}{cc} I & 0 \end{array}\right]$, $x_2 = \left[\begin{array}{cc} q_2^T & \dot{q}_2^T \end{array}\right]^T$,

$$A_2 = \left[\begin{array}{cc} 0 & I \\ -J^{-1}K & -J^{-1}D \end{array}\right], \quad B_2 = \left[\begin{array}{c} 0 \\ J^{-1} \end{array}\right].$$

Note that the matrix A_2 is stable, which is always physically guaranteed. It follows from (3) that

$$\begin{aligned} q_2(t) &= C_2 e^{A_2 t} x_2(0) + \tau' + C_2 e^{A_2 t} * B_2 K q_1 \\ &\quad + C_2 e^{A_2 t} * B_2 P(q_1, \dot{q}_1), \end{aligned} \quad (4)$$

where $*$ denotes the convolution operation and $\tau' = C_2 e^{A_2 t} * B_2 \tau$.

To make the analysis easier to understand, let us proceed the I/O robust control design for two different cases. In the next section, we shall first study the simple case that the parameters K, J, and D are known. The general case that the parameters K, J, and D are unknown will be investigated in the subsequent section.

3 Simplified I/O Control

In this section, I/O robust control design is studied for the case that $J = J_o$, $K = K_o$, and $D = D_o$ for some nominal value matrices J_o, K_o, and D_o. With the knowledge of A_{2o} and B_{2o} where

$$A_{2o} = \left[\begin{array}{cc} 0 & I \\ -J_o^{-1}K_o & -J_o^{-1}D_o \end{array}\right], \quad B_{2o} = \left[\begin{array}{c} 0 \\ J_o^{-1} \end{array}\right],$$

the transition matrix $e^{A_{2o}t}$ can be calculated. Consequently, the intermediate variable

$$\tau_o' = \tau' \,|_{J=J_o, D=D_o, K=K_o} = C_2 e^{A_{2o}t} * B_{2o}\tau$$

is available in control design. In stead of calculating directly $e^{A_{2o}t}$ off-line, one can use the following augmented system to generate τ_o':

$$\dot{z} = A_{2o} z + B_{2o}\tau, \quad \tau_o' = C_2 z, \quad (5)$$

where z is the augmented state of appropriate order. More specifically, letting $z = \left[\begin{array}{cc} z_1^T & z_2^T \end{array}\right]^T$ yields

$$\begin{aligned} \tau_o' &= z_1 \\ \dot{z}_1 &= z_2 \quad (6) \\ \dot{z}_2 &= -J_o^{-1}K_o z_1 - J_o^{-1}D_o z_2 + J_o^{-1}\tau. \quad (7) \end{aligned}$$

It is worth noting here that, although matrices in (3) are known, q_2 or x_2 can not be calculated due to the uncertainty $P(q_1, \dot{q}_1)$.

Substituting solution (4) and (7) into (1) yields the error system:

$$\begin{aligned} \dot{x}_1 &= A_1 x_1 + B_1(\Delta A - \tau_o' + q_1) \\ &= \overline{A} x_1 + B_1(\Delta A' - \tau_o' + q_1), \end{aligned} \quad (8)$$

where $K_p > 0$ is gain matrices chosen by the designer, $\Delta A = K_o^{-1}M(q_1)\ddot{q}_1^d + K_o^{-1}N(q_1, \dot{q}_1) - C_2 e^{A_{2o}t} x_2(0) - C_2 e^{A_{2o}t} * B_{2o} P(q_1, \dot{q}_1)$, $\Delta A' = \Delta A + K_o^{-1}K_p e_1 - C_2 e^{A_{2o}t} * B_{2o} K_o q_1$, $B_1 = \left[\begin{array}{cc} 0 & K_o^T M^{-1} \end{array}\right]^T$, and

$$A_1 = \left[\begin{array}{cc} 0 & I \\ 0 & 0 \end{array}\right], \quad \overline{A} = \left[\begin{array}{cc} 0 & I \\ -M^{-1}(q_1)K_p & 0 \end{array}\right].$$

Input-output robust control design as well as corresponding stability and tracking performance will be investigated in terms of error system (8) and augmented system (5). The analysis is done using Lyapunov's direct method. To apply the Lyapunov technique, let us choose a positive definite function V as a Lyapunov function candidate:

$$V = V_1 + V_2 = x_1^T P x_1 + \frac{1}{2}(z - u)^T (z - u), \quad (9)$$

where u is the vector to be chosen later, $\alpha > 0$ is a constant scalar, and

$$P = \frac{1}{2}\left[\begin{array}{cc} K_p + \alpha^2 M(q_1) & \alpha M(q_1) \\ \alpha M(q_1) & M(q_1) \end{array}\right].$$

The following lemma illustrates the property of the derivative of sub-Lyapunov function $V_1(x_1)$ along every trajectory of system (8). The proof requires only basic matrix calculations and therefore is omitted.

Lemma 1: *Consider system (1) and (2) under Properties P.1 and P.2 and Assumptions A.1 and A.2. Assume in addition that the matrices K, D, and J be constant and known. Then, the sub-Lyapunov function $V_1(x_1)$ satisfies the following inequality:*

$$\begin{aligned} \dot{V}_1(x_1) \leq\ & -\lambda_1 \|e_1\|^2 + \|w(x_1)\| \varrho_1(x_1) - w^T(x_1) K_o (\tau_o' \\ & - q_1 - K_o^{-1} K_p e_1 + C_2 e^{A_{2o}t} * B_{2o} K_o q_1) \end{aligned} \quad (10)$$

where $\lambda_{min}(\cdot)$ and $\lambda_{max}(\cdot)$ represent the operation of taking the minimum and maximum eigenvalues, respectively, $\lambda_1 = \alpha \lambda_{min}(K_p)$, $w(x_1) \triangleq \left[\begin{array}{cc} \alpha I & I \end{array}\right] x_1$, and

$$\begin{aligned} \varrho_1(x_1) &= \overline{m}(q_1)\|\ddot{q}_1^d\| + \rho_1(x_1)\|\dot{q}_1\| + \rho_2(x_1) + \rho_3(x_1) \\ &\quad + \|w(x_1)\| \rho_5(x_1) + \alpha \overline{m}(q_1) \|\dot{e}_1\| \\ &\quad + \|K_o C_2 e^{A_{2o}t}\| \cdot \rho_7 \\ &\quad + \int_{t_0}^{t} \|K_o C_2 e^{A_{2o}(t-s)} B_{2o}\| \rho_4(x_1(s)) ds. \end{aligned}$$

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Remark: The bounding function $\varrho_1(x_1)$ does not depend on internal state x_2 and only requires an estimate (or bound) on the initial condition of x_2.

Upon having Lemma 1, we can proceed input-output robust control design. The proposed I/O controller is generated by the following recursive mapping:

$$\begin{aligned}
u &= \begin{bmatrix} u_1^T & u_2^T \end{bmatrix}^T \\
u_1 &= u_{11} + u_{12} \\
u_{11} &= q_1 + K_o^{-1}K_p e_1 + K_o^{-1}\overline{m}(q_1)w(x_1) \\
&\quad -C_2 e^{A_{2o}t} * B_{2o}K_o q_1 \\
u_{12} &= K_o^{-1}\frac{\mu_1(x_1,t)|\mu_1(x_1,t)|^{\nu_1}}{\left(|\mu_1(x_1,t)|^{\nu_1+1}+\epsilon_1^{1+\nu_1}\right)}\varrho_1(x_1), \\
u_2 &= u_1 - z_1 + K_o^T w(x_1) \\
&\quad +\frac{\mu_2(x_1,t)|\mu_2(x_1,t)|^{\nu_2}}{\left(|\mu_2(x_1,t)|^{\nu_2+1}+\epsilon_2^{1+\nu_2}\right)}g_2(x_1,t), \\
\tau &= J_o u_2 + J_o u_1 + (K_o - J_o)z_1 + (D_o - J_o)z_2 \\
&\quad +J_0\frac{\mu_3(x_1,t)|\mu_3(x_1,t)|^{\nu_i}}{\left(|\mu_3(x_1,t)|^{\nu_i+1}+\epsilon_3^{1+\nu_i}\right)}g_3(x_1,t), \quad (11)
\end{aligned}$$

where $\epsilon_j > 0$ and $\nu_j \geq 0$, $j = 1,2,3$, are constants, the constants ν_j are chosen such that the first order partial derivatives of u_j with respect to their variables are well-defined, $|||\cdot|||$ denotes a known bounding function on the magnitude of the argument, and

$$\mu_1(x_1,t) = w(x_1)\varrho_1(x_1),$$

$$g_2(x_1,t) = |||\dot{u}_1|||, \qquad \mu_2(x_1,t) = (u_1 - z_1)g_2(y,u,t),$$

$$g_3(x_1,t) = |||\dot{u}_2|||, \qquad \mu_3(x_1,t) = (u_2 - z_2)g_3(x_1,t).$$

The recursive mapping basically involves finding bounding functions of $|\dot{u}_1|$ and $|\dot{u}_2|$ for obtaining u_2 and τ, respectively. The terms $|\dot{u}_1|$ and $|\dot{u}_2|$ can be bounded by first developing bounds for the first-order partial derivatives of u_i with respect to its variables, and then by determining the bounds for the first-order time derivatives of its variables using dynamic equations (5) and (8). The bounding functions are usually obtained by taking Euclidean norm. There are several ways to guarantee differentiability of u_1 and u_2. The easiest way is to use some user-defined norm such as $\|y\|_u \triangleq \sqrt{\|y\|^2 + \varepsilon}$ for some $\varepsilon > 0$. It is worth mentioning here that $\dot{u}_1$ will contain $\dot{x}_1$ which in turn depends on x_2. However, the bounding function on $|\dot{u}_1|$ will only depend on x_1 and τ_o' since x_2 can be bounded using (4) with $\tau' = \tau_o'$.

We are now in a position to state the following theorem on control (11) which requires only input-output measurements and is for now robust with respect to any bounded uncertainties except the parameter variation of J, D, and K. Its proof is omitted for briefness since it can be done in a similar fashion as that in [14].

Theorem 1: *Consider system (1) and (2) under Properties P.1 and P.2 and Assumptions A.1 and A.2. Assume in addition that the matrices K, D, and J be constant and known. If I/O robust controller τ is chosen to be the outcome of the mapping procedure (11), then the trajectory tracking error x_1 (both e and $\dot{e}$) is globally and uniformly ultimately bounded. That is, as time approaches infinity, the magnitude of the tracking error characterized by $V_1(e,\dot{e})$ becomes no larger than the design parameter ϵ^* where*

$$\frac{1}{\min\{\frac{2\alpha\lambda_{min}(K_p)}{\lambda_{max}(K_p)}, 2\}}\sum_{i=1}^{3}\epsilon_i.$$

Furthermore, the robust control τ is continuous and globally, uniformly bounded.

4 Robust I/O Control

In this section, input-output control design is proceeded without knowledge of matrices J, D, and K. When these matrices are not known, the intermediate variable τ' defined in (4) can not be calculated. However, it is noted that, if a nominal augmented system similar to (5) can be obtained by defining τ_o' properly, most of the analysis in the previous section follows. Therefore, we shall focus on how to generate from (3) an equation similar to (5) which depends only on nominal values.

Consider the following linear time-varying system:

$$F(s,t)y = v + d, \qquad (12)$$

where y, v and d are column vectors of n dimension, y is the output, v is the corresponding input, d is disturbance, $F(s,t)$ is the so-called time-varying differential operator defined by $F(s,t) = Js^2 + Ds + K$. That is, system (12) can be rewritten in terms of differential equation as $J\ddot{y} + D\dot{y} + Ky = v + d$. More details on time-varying differential operator can be found in [20, 7].

The problem to be studied is under what input v the closed-loop system becomes

$$F_o(s)J_o^{-1}Jy = r + d', \qquad (13)$$

where J_o, D_o, and K_o are constant matrices represents any choice of nominal values of J, D, K respectively, r is any reference input, d' is the equivalent disturbance, and differential operator $F_o(s)$ is given by $F_o(s) = J_o s^2 + D_o s + K_o$.

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Let us introduce the following auxiliary signals:

$$\dot{\eta}_1 = -\eta_1 + G_1 v, \tag{14}$$
$$\dot{\eta}_2 = -\eta_2 + G_2 y, \tag{15}$$

where η_1 and η_2 are n-th order column vectors, initial conditions of η_1 and η_2 are assumed without loss of any generality to be zero, and G_1 and G_2 are n-th order diagonal matrices.

The input v is selected as

$$v = G_3 y + r + \eta_1 + \eta_2, \tag{16}$$

where G_3 is also an n-th order diagonal matrix. Utilizing differential operator, we can rewrite the input v as $v = G_3 y + r + (s+1)^{-1}[G_1 v + G_2 y]$, or

$$v = (sI + I - G_1)^{-1}\{[(s+1)G_3 + G_2]y + (s+1)r\}.$$

Substituting the above equation into (12) yields $[(sI + I - G_1)(Js^2 + Ds + K) - (s+1)G_3 - G_2]y = (s+1)r + (sI + I - G_1)d$. It then follows that the closed loop system is given by (13) with $d' = (s+1)^{-1}(sI + I - G_1)d$ iff the following identity holds:

$$[(sI + I - G_1)(Js^2 + Ds + K) - (s+1)G_3 - G_2] = (s+1)(J_o s^2 + D_o s + K_o)J_o^{-1} J. \tag{17}$$

Since all matrices are diagonal, the above equality can be rewritten as $[(s + 1 - G_{1i})(J_i s^2 + D_i s + K_i) - (s+1)G_{3i} - G_{2i}] = (s+1)(J_{oi} s^2 + D_{oi} s + K_{oi})J_i / J_{oi}$, where subscript i denotes the i-th element on the diagonal. It is easy to verify that the above equality holds if G_1, G_2, and G_3 are chosen such that

$$\begin{aligned} G_{1i} &= D_i/J_i - 2\dot{J}_i/J_i - D_{oi}/J_{oi}, \\ G_{3i} &= \dot{D}_i + K_i + (1 - G_{1i})D_i - 3\ddot{J}_i - 2\dot{J}_i \\ &\quad -2D_{oi}\dot{J}_i/J_{oi} - K_{oi}J_i/J_{oi} - D_{oi}J_i/J_{oi}, \\ G_{2i} &= \dot{K}_i + (1 - G_{1i})K_i - \dot{G}_{3i} - d^3 J_i/dt^3 \\ &\quad -(1 + D_{oi}/J_{oi})\dddot{J}_i - (K_{oi} + D_{oi})\ddot{J}_i/J_{oi} \\ &\quad -K_{oi}\dot{J}_i/J_{oi}. \end{aligned} \tag{18}$$

Solution (18) guarantees identity (17) which in turn guarantees that the closed-loop system is internally stable. The above relations will be used to determine known bounds on G_1, G_2, and G_3. These bounds are required in designing robust control.

It should be emphasized here that the introduction of differential operators implies zero initial conditions. It is noted that systems given by (3) and (12) are both uniformly asymptotically stable. Consequently, nonzero initial conditions contribute only exponentially decaying terms with which a similar Lyapunov stability proof can be proceeded to yield the same stability results.

Now, reconsider dynamic equation (2) by rewriting as

$$\begin{aligned} & J\ddot{q}_2 + D\dot{q}_2 + Kq_2 \\ = \;& \tau + Kq_1 + P(q_1, \dot{q}_1) \\ = \;& [G_3 q_2 + \tau + \eta_1 + \eta_2] + [Kq_1 + P(q_1, \dot{q}_1) - G_3 q_2 \\ & -\eta_1 - \eta_2] \triangleq v + d, \end{aligned}$$

where η_1 and η_2 are given by (14) and (15) with $r = \tau$ and $y = q_2$, and G_i are given by the solution (18). It follows from the discussion from (12) to (18) that dynamics (2) becomes

$$\ddot{q}_2' + J_o^{-1} D_o \dot{q}_2' + J_o^{-1} K_o q_2' = \tau + P'(q_1, \dot{q}_1, \zeta) \tag{19}$$

where $q_2' = Jq_2$, $\zeta_1 = e^{-t} * \tau$, $\zeta_2 = e^{-t} * \zeta_2$, $\zeta_3 = e^{-t} * \zeta_3$, $\zeta = \begin{bmatrix} \zeta_1^T & \zeta_2^T & \zeta_3^T \end{bmatrix}^T$, $P'(q_1, \dot{q}_1, \zeta) = (s+1)^{-1}(sI + I - G_1)[Kq_1 + P(q_1, \dot{q}_1) - G_3 q_2 - \eta_1 - \eta_2]$.

Hence, the solution for q_2 is given by

$$\begin{aligned} q_2(t) &= J^{-1} C_2 e^{A_{2o}t} x_2'(0) + J^{-1}\tau_o'' \\ &\quad + J^{-1} C_2 e^{A_{2o}t} * B_{2o} J_o P'(q_1, \dot{q}_1, \zeta), \end{aligned} \tag{20}$$

where $\tau_o'' = C_2 z'$, $z' = \begin{bmatrix} (z_1')^T & (z_2')^T \end{bmatrix}^T$ $x_2'(0) = \begin{bmatrix} q_2^T(t_0)J(t_0) & \dot{q}_2^T J(t_0) + q_2^T(t_0)\dot{J}(t_0) \end{bmatrix}^T$, and

$$\dot{z}' = A_{2o} z' + B_{2o} J_o \tau, \tag{21}$$

which is similar to (5).

There are two differences between solutions (20) and (4). First, τ_o'' should be used to replace τ_o'. Second, the "total" uncertainty $P'(q_1, \dot{q}_1, \zeta)$ is far more complicated than $P(q_1, \dot{q}_1)$. Despite of the differences, the two solutions share the same type of expressions. As a consequence, analysis can be proceeded to yield the same kind of stability results by duplicating those in the preceding section with minor modifications.

To avoid redundancy, we shall omit those derivations which are essentially the same but only include two necessary changes. The first change is that the bounding function $\varrho_1(x_1)$ needs to be changed. It follows from (14), (15), and (16) that $(sI + I - G_1)\eta_1 = G_1 G_3 q_2 + G_1 \tau + G_1 \eta_2$ and $(s+1)\eta_2 = G_2 q_2$. Hence, we have

$$\begin{aligned} & P'(q_1, \dot{q}_1, \zeta) \\ = \;& Kq_1 + P(q_1, \dot{q}_1) - G_3 q_2 - (s+1)^{-1}[G_1 K q_1 \\ & + G_1 P(q_1, \dot{q}_1) - G_2 q_2] - (s+1)^{-1} G_1 \tau \\ = \;& Kq_1 + P(q_1, \dot{q}_1) - G_3 q_2 - (s+1)^{-1}[G_1 K q_1 \end{aligned}$$

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$$
\begin{aligned}
&+G_1P(q_1,\dot{q}_1)-G_2q_2]-G_1\zeta_1+(s+1)^{-1}\dot{G}_1\zeta_1\\
=\;& Kq_1+P(q_1,\dot{q}_1)-G_3q_2-(s+1)^{-1}[G_1Kq_1\\
&+G_1P(q_1,\dot{q}_1)-G_2q_2]-(s+1)^2[G_1\zeta_3\\
&-3(s+1)^{-1}\dot{G}_1\zeta_3+3(s+1)^{-2}\ddot{G}_1\zeta_3\\
&-(s+1)^{-3}\frac{d^3G_1}{dt^3}\zeta_3\Big],
\end{aligned}
$$

in which the last two steps are derived by integration by part. Such steps are necessary in order to find a well-defined robust control τ by guaranteeing that the second-order time derivative of the bounding function to be determined does not depend on τ explicitly. The new bounding function $\varrho_1'(x_1,\zeta_2,\zeta_3)$ to replace $\varrho_1(x_1)$ in Lemma 1 can then be chosen to be

$$
\begin{aligned}
&\varrho_1'(x_1,\zeta_2,\zeta_3)\\
=\;& \|K_pe_1\|+\overline{m}(q_1)\|\ddot{q}_1^d\|+\rho_1(x_1)\|\dot{q}_1\|+\rho_2(x_1)\\
&+\rho_3(x_1)+\|w(x_1)\|\rho_5(x_1)+\alpha\overline{m}(q_1)\|\dot{e}_1\|\\
&+\|\underline{J}\|^{-1}\left[\|C_2e^{A_{2o}t}\|\cdot\|\overline{J}\|\rho_7+|||G_1|||\cdot\|\zeta_3\|\right.\\
&+\int_{t_0}^{t}\|C_2e^{A_{2o}(t-s)}B_{2o}J_o\|(\|\overline{K}\|||q_1\|+\rho_4(x_1)\\
&\left.+|||G_3|||\cdot|||q_2|||+3|||\ddot{G}_1|||\cdot\|\zeta_3\|)ds\right.\\
&+3\int_{t_0}^{t}\|C_2'e^{A_{2o}(t-s)}B_{2o}J_o\|||G_1|||\cdot\|\zeta_3\|ds\\
&+3\int_{t_0}^{t}\|C_2''e^{A_{2o}(t-s)}B_{2o}J_o\|||\dot{G}_1|||\cdot\|\zeta_3\|ds\\
&+\int_{t_0}^{t}\|C_3e^{A_3(t-s)}B_3\|(\|\overline{K}\|\cdot|||G_1|||\cdot\|q_1\|+|||G_1|||\\
&\times\rho_4(x_1)+|||G_2|||\cdot|||q_2|||+\left\|\left|\frac{d^3G_1}{dt^3}\right|\right\|\cdot\|\zeta_3\|)ds\Big],
\end{aligned}
$$

where $C_2' = \left[\begin{array}{cc} I-J_o^{-1}K_o & 2I-J_o^{-1}D_o \end{array}\right]$, $C_2'' = \left[\begin{array}{cc} I & I \end{array}\right]$, $B_3=\left[\begin{array}{ccc} 0 & 0 & I \end{array}\right]^T$, $C_3=\left[\begin{array}{ccc} I & 0 & 0 \end{array}\right]$,

$$
A_3=\left[\begin{array}{ccc} 0 & I & 0\\ 0 & 0 & I\\ -J_o^{-1}K_o & -J_o^{-1}(K_o+D_o) & -I-J_o^{-1}D_o \end{array}\right],
$$

and $|||q_2|||$ denotes a known bounding function on $\|q_2\|$ which can be determined from (4). That is,

$$
\begin{aligned}
&\|q_2\|\\
\le\;& \|C_2e^{A_2t}\|\rho_7+\|C_2e^{A_2t}*B_2\tau\|\\
&+\|C_2e^{A_2t}*B_2Kq_1\|+\|C_2e^{A_2t}*B_2P(q_1,\dot{q}_1)\|\\
\le\;& |||C_2e^{A_2t}|||\rho_7+\|\underline{J}\|^{-1}\cdot\|\zeta_2\|\\
&+\int_{t_0}^{t}|||C_2'''e^{A_2(t-s)}B_2|||\cdot\|\zeta_2(s)\|ds\\
&+\int_{t_0}^{t}|||C_2e^{A_2(t-s)}B_2K|||\cdot\|q_1(s)\|ds\\
&+\int_{t_0}^{t}|||C_2e^{A_2(t-s)}B_2|||\cdot\rho_4(x_1(s))ds,
\end{aligned}
$$

where $C_2''' = \left[\begin{array}{cc} 2J^{-1}-J^{-1}D & J^{-1}-J^{-1}K \end{array}\right]$.

The second modification is that the mapping for generating I/O control should be re-designed to be the following:

$$
\begin{aligned}
u &= \left[\begin{array}{cc} u_1^T & u_2^T \end{array}\right]^T\\
u_1 &= u_{11}+u_{12}\\
u_{11} &= q_1+\overline{J}\underline{K}^{-1}\overline{m}(q_1)w(x_1)\\
u_{12} &= \overline{J}\underline{K}^{-1}\frac{\mu_1(x_1,t)|\mu_1(x_1,t)|^{\nu_1}}{\left(|\mu_1(x_1,t)|^{\nu_1+1}+\epsilon_1^{1+\nu_1}\right)}\varrho_1(x_1,\zeta_3),\\
u_2 &= u_1-z_1'+\frac{\mu_2(x_1,t)|\mu_2(x_1,t)|^{\nu_2}}{\left(|\mu_2(x_1,t)|^{\nu_2+1}+\epsilon_2^{1+\nu_2}\right)}g_2(x_1,t),\\
\tau &= u_2+u_1+(J_o^{-1}K_o-I)z_1'+(J_o^{-1}D_o-I)z_2'\\
&\quad+\frac{\mu_3(x_1,t)|\mu_3(x_1,t)|^{\nu_i}}{\left(|\mu_3(x_1,t)|^{\nu_i+1}+\epsilon_3^{1+\nu_i}\right)}g_3(x_1,t), \qquad (22)
\end{aligned}
$$

where $\epsilon_j > 0$ and $\nu_j \ge 0$, $j=1,2,3$, are constants, the constants ν_j are chosen such that the first order partial derivatives of u_j with respect to their variables are well-defined, $\mu_1(x_1,t)=w(x_1)\varrho_1(x_1,\zeta_3)$, $g_2(x_1,t)=|||\dot{u}_1|||+\|\underline{J}\|^{-1}\|\overline{K}\|||w(x_1)\|$, $\mu_2(x_1,t)=(u_1-z_1')g_2(y,u,t)$, $g_3(x_1,t)=|||\dot{u}_2|||$, and $\mu_3(x_1,t)=(u_2-z_2')g_3(x_1,t)$.

By having made the two modifications and by using similar arguments as those for Lemma 1 and Theorem 1, we can establish the following results on robust input-output control of manipulators with flexible joints.

Lemma 2: *Consider system (1) and (2) under Properties P.1 and P.2 and Assumptions A.1, A.2, and A.3. The sub-Lyapunov function $V_1(x_1)$ satisfies the following inequality:*

$$
\begin{aligned}
\dot{V}_1(x_1) \le\;& -\lambda_1\|e_1\|^2+\|w(x_1)\|\varrho_1'(x_1,\zeta_2,\zeta_3)\\
&-w^T(x_1)K(\tau_o''-q_1), \qquad (23)
\end{aligned}
$$

where λ_1 is defined in Lemma 1.

Theorem 2: *Consider system (1) and (2) under Properties P.1 and P.2 and Assumptions A.1, A.2, and A.3. If I/O robust controller τ is chosen to be the outcome of the mapping procedure (22), then the trajectory tracking error x_1 (both e and $\dot{e}$) is globally and uniformly ultimately bounded. That is, as time approaches infinity, the magnitude of the tracking error characterized by $V_1(e,\dot{e})$ becomes no larger than the design parameter ϵ^* given in Theorem 1. Furthermore, the robust control τ is continuous and globally, uniformly bounded.*

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Conclusion

A robust control design method was presented in this paper for robot manipulators with flexible joints. In constrast to existing robust control results and to other design techniques such as adaptive control and singular perturbation, the proposed control guarantees global stability while using only measurement of joint position and velocity. Because acceleration and jerk measurement needed to apply any other techniuqes is no longer required, the proposed control is more practical and attractive. Furthermore, the proposed control works robustly to compensate any unknown but bounded uncertainties including unknown flexibility, (time-varying) parameters, load variation, and disturbances. Regardless of uncertainties, position and velocity tracking errors have been shown to satisfy any arbitrarily small tracking accuracy requirement.

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